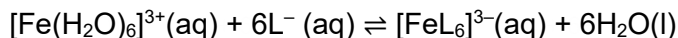


- 29** A stability constant is an equilibrium constant for the formation of a complex relative to its aqua complex in solution.

Consider the formation of $[\text{FeL}_6]^{3-}(\text{aq})$ in solution:



where L^{-} can be either F^{-} or CN^{-} .

The equilibrium constant, K_{stab} , of the above reaction is

$$K_{\text{stab}} = \frac{[\text{FeL}_6^{3-}]}{[\text{Fe}(\text{H}_2\text{O})_6^{3+}][\text{L}^{-}]^6}$$

The table below shows the values of the stability constants and colours of some iron complexes.

complex	K_{stab}	colour
$[\text{FeF}_6]^{3-}$	2×10^{15}	colourless
$[\text{Fe}(\text{CN})_6]^{3-}$	1×10^{31}	orange red

What would be observed when $\text{NaCN}(\text{aq})$ is added to a solution of $\text{Fe}(\text{NO}_3)_3$, followed by adding $\text{NaF}(\text{aq})$?

- A** The yellow solution turns orange red and remains orange red.
- B** The yellow solution turns orange red and then to colourless.
- C** The green solution turns orange red and remains orange red.
- D** The green solution turns orange red and then to colourless.